Ziru Wei

ziru.wei@colorado.edu | ziru-wei.github.io

Keywords: Human-Computer Interaction; Human-Robot Interaction; Context-Aware Interaction

Research Interests

My research explores AI interfaces out of screen to support and augment humans undisruptively in everyday physical environments. I develop medium-agnostic and context-adaptive systems spanning modalities from extended reality to robotic interfaces, and investigate how humans perceive them.

Education

The ATLAS Institute, University of Colorado Boulder Starting Fall 2026

- PhD in Creative Technology and Design
- Advisor: Prof. Ryo Suzuki

Carnegie Mellon University Graduated May 2026

- Master of Science in Computational Design (research-based)
- Advisor: Prof. Alexandra Ion
- Thesis Project: *Toward Interacting with Proactive Intelligence in Everyday Physical Environments*

Soochow University Graduated Jun 2024

- Bachelor of Architecture

Publications

[P3] **While You're Stil Busy: Context-Aware Approach Optimization for Proactive Mobile Robots**
Ziru Wei, Di Wen, Leyuan Chen, Hassan Shahzad, Alexandra Ion *UIST'26* (ACM Symposium on User Interface Software and Technology, 2026), full paper accepted

[P2] **Objects with Arms: How Task-Relevant Embodiment Shapes Perception in Physical Collaboration**
Violet Yinuo Han, Ziru Wei, Aiden Yiliu Li, Chris Wu, Alexandra Ion *RO-MAN'26* (IEEE International Conference on Robot and Human Interactive Communication, 2026), full paper accepted

[P1] **On-site Holographic Building Construction: A Case Study of Aurora**
Sijie Liu, Ziru Wei, Sining Wang. *CAADRIA'22* (Proceedings of the Association for Computer-Aided Architectural Design Research in Asia, 2022) full paper accepted (~30% acceptance, top-tier conference in computational design)

Research Experiences

Master's Student | Interactive Structures Lab, Carnegie Mellon University Mar 2025 - Present

- Advisor: Prof. Alexandra Ion
- Developed software and hardware systems for proactive robotic interfaces that blended into everyday environments, and understood human perceptions of them (See P2, P3).

Research Intern | WHY Research Lab, School of Architecture, Carnegie Mellon University Aug 2024 - Jan 2025

- Advisor: Prof. Daragh Byrne
- Replicated an e-waste scanner by disassembling and re-soldering disk drives, integrating a Raspberry Pi controller; built the WasteStation database in Notion to map components and their potential reuse applications

Research Intern | Humachine Lab, Architecture Department, Soochow University May 2021 - Aug 2022

- Advisor: Prof. Sining Wang
- Designed MR workflows and four on-site collaboration methods for nonlinear façade assembly with limited devices and a small construction team; documented the design-to-construction process and contributed to the literature review part in paper writing (See P1)

Awards and Honors

Computational Design Commendation, Carnegie Mellon University 2025

Computational Design Commendation, Carnegie Mellon University 2024

Merit Scholarship, Carnegie Mellon University 2024

Outstanding Undergraduate Thesis (Top 1%), Soochow University 2024

Excellence Award, Shanghai Youth Architectural Design Competition 2023

Innovation & Academic Excellence Scholarship, Soochow University 2020 - 2022

Overall Excellence Award Winner, Solar Decathlon China 2022

First Prize (Top 2%) in "Zijin Award" of Architectural Design Contest 2022

METTLER TOLEDO Scholarship (Top 2%) 2019

Mentoring

Tanvi Handoo, Undergraduate Student in Computer Science, Carnegie Mellon University May 2025 - Sept 2025

Activities

Course Guest Reviewer, Carnegie Mellon University Jan 2025 - May 2025

Student Volunteer, des[AI]gn conference 2024, American Institute of Architecture Students Oct 2024

Skills

Technical
- Hardware: Arduino, Raspberry Pi
- Programming: Python, PyTorch, Rstudio, C#, Pascal, HTML, CSS, JavaScript

Design & Production
- Software: Unity, Rhino, Grasshopper with GHPython, Blender, AutoCAD, Adobe Creative Suite, Figma, Procreate
- Fabrication: 3D printing, Soldering and electronic wiring, Welding (basic), Woodworking

Languages
- English (Fluent), Mandarin (Native), Portuguese (Beginner)